

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

DATABASE MANAGEMENT SYSTEMS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

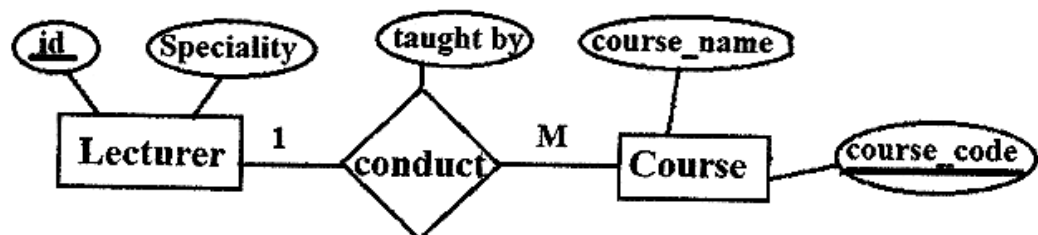
All questions are compulsory.

1. (a) Enlist and explain various abstraction level of database. 5(CO1)
 (b) For the given relation 'Person' perform the operation :

Id	Name	Address	Hobby
1123	John	123 Main	stamp
1123	John	123 Main	coin
5556	Mary	7 Lake Dr	hiking
9876	Bart	5 Pine St	stamp

- (i) $\pi_{\text{Name, Address}}(\text{Peron})$.
 (ii) $\pi_{\text{Id, Name}}(\sigma_{\text{Hobby} = \text{'stamps'} \text{ OR } \text{Hobby} = \text{'coins'}}(\text{Person}))$. 5(CO1)

2. (a) Convert the following ER Model into relational model.



5CO4)

- (b) For the given relation $R(A, B, C, D, E, F)$ with FDs,
 $\{A \rightarrow BC, B \rightarrow C, A \rightarrow B, AB \rightarrow C\}$, determine the minimal cover of
 FD set. 5(CO1)
3. Design a database considering the following database schema –
 Employee(ename, street, city)
 Works(ename, company_name, salary)
 Company(company_name, city)
 Write Relational algebra expression AND SQL queries for the following :—
- (a) Retrieve name of all employees who works for the company "ABC".
- (b) Retrieve name, street and city employees who are working for the company "ABC".
- (c) Find the name and salary of the person(s) who work for the company "PQR" located in the city "Nagpur".
- (d) Find the company name, city and salary where the company offers maximum salary.
- (e) Find number of employees who are getting salary between Rs. 50,000 to Rs. 1,00,000. 10(CO3,5)
4. (a) Explain various types of File organizations in DBMS. 5(CO2)
- (b) Following key values are to stored in hash table :
 Keys : 3, 2, 9, 6, 11, 13, 7, 12.
 Consider a hash table of size ten, with starting index zero, and a hash function, $h(K) = 2K + 3$. Assuming the hash table is initially empty, determine the contents of the table when the given key values are inserted into the table using division method and chaining method of hashing. 5(CO2)
5. (a) What is query processing ? Explain the steps involved in query processing and explain it in detail. 5(CO1)

- (b) Explain ACID property of transaction in DBMS. 5(CO1)
6. (a) Let S be the following schedule of operation of three transactions T1, T2 and T3 in relational database system :
S : R2(Y), R1(X), R2(Z), R1(Y), W1(X), R2(Z), W2(Y), R3(X), W3(Z).
Verify whether schedule S is conflict serializable or not. 5(CO5)
- (b) What is deadlock ? Explain the different deadlock prevention schemes. 5(CO1)

